

Wavelet-Supervised Test-Time Prototype Adaptation for Zero-Shot Anomaly Detection

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Abstract. Zero-shot industrial anomaly detection requires image-level anomaly recognition and pixel-level anomaly localization without target-category training images or anomaly annotations. Existing CLIP-based methods usually rely on fixed normal/abnormal text prototypes, which cannot fully capture instance-specific defect appearances and may confuse true defects with normal material textures or structural boundaries. We propose Wavelet-Supervised Test-Time Prototype Adaptation (WPTA), which uses Haar wavelet cues as patch evidence reliability for test-time prototype adaptation rather than as an additive anomaly map. WPTA freezes CLIP, constructs boundary-aware wavelet reliability on the CLIP patch feature grid, combines it with the initial semantic anomaly prior to select visual normal/abnormal anchors, and calibrates prototypes through confidence-gated conservative updates. Controlled ablations on MVTec and VisA show that direct wavelet fusion degrades localization, while semantic-spectral evidence selection and conservative calibration improve test-time prototype adaptation. Across five industrial benchmarks, the calibrated inference system improves the fixed AnomalyCLIP baseline by +1.0/+4.5/+3.4/+2.8 on average in P-AUROC/P-AUPRO/I-AUROC/I-AP.

Keywords: Zero-shot anomaly detection · CLIP · Test-time adaptation · Wavelet reliability

1 Introduction

Zero-shot anomaly detection (ZSAD) aims to detect and localize anomalies on unseen target categories without using target-domain training images. CLIP-based ZSAD methods have made this setting practical by comparing visual features with normal and abnormal text prototypes. However, a fixed abnormal prototype only provides a generic anomaly direction. It does not necessarily match the concrete defect appearance in the current image, such as subtle scratches, missing edges, surface contamination, or texture damage. As a result, genuine defects may be underestimated, while normal boundaries, reflections, or material textures may be activated as false positives.

Frequency and wavelet cues provide useful but risky complementary information. Many industrial defects manifest as local high-frequency disturbances, and Haar wavelets can separate high-frequency details from low-frequency structures on feature grids. Nevertheless, high-frequency response is not equivalent to

anomaly evidence. Normal object contours, reflective surfaces, and fine-grained textures can also produce strong wavelet responses. In controlled tests, direct wavelet fusion lowers pixel-level (AUROC, AUPRO) on MVTec from (91.2, 83.2) to (88.7, 80.4), and on VisA from (95.5, 86.7) to (94.6, 85.1).

We therefore reposition wavelet cues from final-map scoring to evidence reliability for test-time prototype adaptation. WPTA first computes an initial semantic anomaly prior S_0 using fixed normal/abnormal prototypes. It then applies Haar decomposition to the CLIP patch feature grid, estimates high-frequency texture energy and low-frequency structural edges, and constructs a boundary-aware reliability map. The semantic prior S_0 provides the anomaly direction, while the wavelet reliability W constrains whether local evidence is trustworthy. The selected visual anchors are used to conservatively calibrate prototypes without backpropagation or CLIP parameter updates.

Our evidence is organized at two levels. First, controlled MVTec/VisA ablations validate the WPTA mechanism: semantic-only prototype adaptation improves over fixed prototypes, wavelet-guided reliability further improves localization, and conservative update gives additional gains. Second, five industrial benchmarks report the system-level effect of the calibrated inference stack. Since the five-dataset final systems use dataset-specific calibration modules, we explicitly separate system-level gains from the causal evidence for WPTA.

The contributions are summarized as follows:

- We propose WPTA, which turns Haar wavelet cues into patch evidence reliability for test-time prototype adaptation rather than direct final-map fusion.
- We design semantic-spectral evidence selection, jointly using the semantic anomaly prior and boundary-aware wavelet reliability to construct visual normal/abnormal anchors.
- We perform conservative, training-free prototype calibration with frozen CLIP and validate the mechanism through controlled MVTec/VisA ablations.
- We report five industrial benchmark results with a configuration audit that separates calibrated-system gains from WPTA mechanism evidence.

2 Related Work

CLIP-based zero-shot anomaly detection.

CLIP [11] provides a general image–text alignment foundation for open vocabulary recognition. WinCLIP [7] performs window-level CLIP matching for zero-/few-shot anomaly classification and segmentation. AnomalyCLIP [13] learns object-agnostic normal/abnormal prompts, while AdaCLIP [4] adapts CLIP with hybrid learnable prompts. CLIP-AD [5] improves zero-shot anomaly detection with language-guided staged dual-path modeling. These methods show the value of CLIP semantic prototypes, but most of them use fixed or trained prototypes without training-free calibration from local evidence in the current test image.

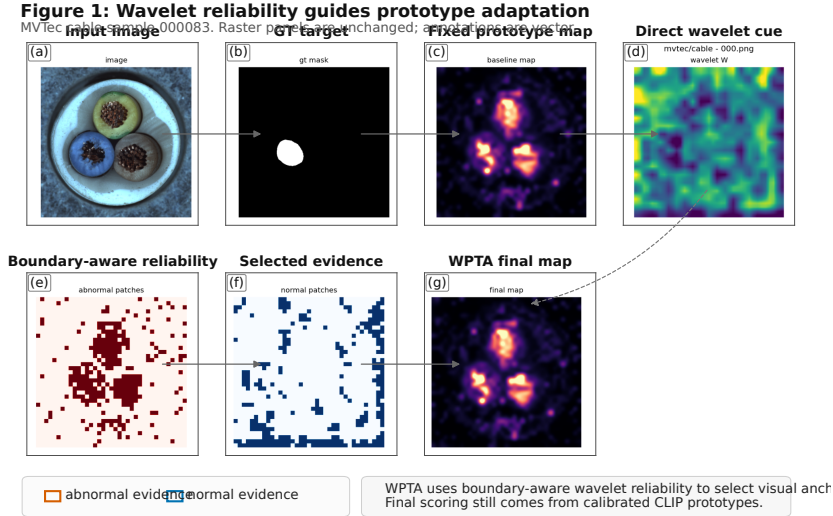


Fig. 1. Motivating example. Wavelet cues are useful as evidence reliability for prototype adaptation rather than as direct anomaly scores.

Test-time adaptation.

Test-time adaptation uses information from the test sample to mitigate distribution shifts. TPT [12] performs test-time prompt tuning for vision-language models through confidence selection and view consistency. Anomaly detection is more challenging because abnormal regions are sparse, unknown, and often adjacent to normal boundaries or textures. WPTA avoids updating CLIP parameters or adapting from the whole image. It aggregates only patch evidence jointly supported by the semantic prior and wavelet reliability.

Frequency and wavelet cues.

Wavelet representations decompose low-frequency structures and high-frequency details at multiple resolutions [6, 9]. In industrial anomaly localization, high-frequency disturbances may indicate scratches, cracks, or texture damage. However, direct frequency scoring is vulnerable to normal boundaries and textures. WPTA limits wavelet cues to evidence reliability and leaves final anomaly decisions to calibrated CLIP prototypes.

3 Method

3.1 Problem Setting and Overall Framework

Given a test image x , ZSAD predicts a pixel-level anomaly map M and an image-level anomaly score s_{img} . The frozen CLIP image encoder extracts patch features

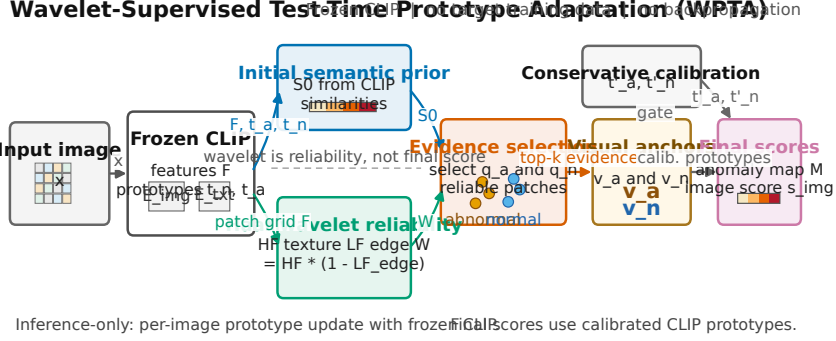


Fig. 2. Overall architecture of WPTA. Wavelet cues supervise evidence reliability for test-time prototype adaptation rather than directly producing the final anomaly map.

$\{f_i\}_{i=1}^N$, and the text encoder provides a normal prototype t_n and an abnormal prototype t_a . WPTA performs inference without target-domain training images, anomaly annotations, backpropagation, or CLIP parameter updates.

The method has four steps, as shown in Fig. 2. First, initial semantic scoring computes a patch-level anomaly prior S_0 . Second, boundary-aware Haar wavelet reliability estimates a reliability map W on the CLIP patch feature grid. Third, semantic-spectral evidence selection aggregates visual normal/abnormal anchors. Fourth, conservative prototype calibration recomputes anomaly probabilities using calibrated prototypes.

3.2 Initial Semantic Anomaly Score

For each patch feature f_i , WPTA computes the initial semantic anomaly prior by a softmax over similarities to normal and abnormal prototypes:

$$S_0(i) = \frac{\exp(\text{sim}(f_i, t_a)/\tau)}{\exp(\text{sim}(f_i, t_a)/\tau) + \exp(\text{sim}(f_i, t_n)/\tau)}. \quad (1)$$

Here $\text{sim}(\cdot, \cdot)$ denotes normalized similarity and τ is the temperature. S_0 provides the semantic direction but is not treated as the final calibrated score.

3.3 Boundary-Aware Haar Wavelet Reliability

WPTA reshapes patch features into $F \in \mathbb{R}^{H \times W \times C}$ and applies one-level Haar DWT to obtain LL , LH , HL , and HH components. The high-frequency texture energy is

$$HF(i) = \frac{1}{C} \sum_{c=1}^C (|LH_i^c| + |HL_i^c| + |HH_i^c|). \quad (2)$$

Since high-frequency energy can also arise from normal structural boundaries, WPTA estimates a low-frequency structural edge response LF_{edge} from LL and constructs

$$W(i) = \text{norm}(HF(i)) \cdot (1 - \text{norm}(LF_{\text{edge}}(i))). \quad (3)$$

Thus, high-frequency texture disturbance increases evidence reliability, while low-frequency structural edges suppress pseudo evidence.

3.4 Semantic-Spectral Evidence Selection

Abnormal evidence should be semantically close to the abnormal prototype and supported by reliable local disturbance. Normal evidence should be close to the normal prototype and not contaminated by abnormal textures or structure-boundary responses. We write the conceptual evidence weights as

$$q_a(i) = S_0(i)^\gamma \cdot \rho(W(i)), \quad q_n(i) = (1 - S_0(i))^\gamma \cdot \rho(1 - W(i)). \quad (4)$$

Implementation uses top- k selection, confidence thresholds, normalization, and confidence gates to control evidence quality.

3.5 Visual Anchors and Conservative Calibration

Selected evidence patches are aggregated into visual anchors:

$$v_a = \frac{\sum_i q_a(i) f_i}{\sum_i q_a(i)}, \quad v_n = \frac{\sum_i q_n(i) f_i}{\sum_i q_n(i)}. \quad (5)$$

The prototypes are then conservatively calibrated:

$$t'_a = \text{Norm}((1 - \alpha)t_a + \alpha v_a), \quad t'_n = \text{Norm}((1 - \beta)t_n + \beta v_n). \quad (6)$$

The update strengths are controlled by confidence gates. In the default controlled setting, $\alpha_0 = 0.0$ and $\beta_0 = 0.01$, so abnormal evidence mainly gates the update, while the actual prototype displacement mainly affects the normal prototype. The final anomaly probability is recomputed by replacing t_a, t_n with t'_a, t'_n in the semantic scoring equation.

4 Experiments

4.1 Experimental Setup

Datasets and Metrics

We evaluate on five industrial anomaly detection benchmarks: MVTec AD [3, 2], VisA [14], MPDD [8], BTAD [10], and DTD-Synthetic [1]. MVTec and VisA are used for controlled WPTA mechanism ablations. The five-dataset table reports system-level results of the calibrated inference stack.

We report Pixel-AUROC, Pixel-AUPRO, Image-AUROC, and Image-AP. Pixel-AUPRO is emphasized because it better reflects region-level anomaly localization quality. All results are deterministic reports without multi-seed confidence intervals or significance tests.

Implementation Details All experiments use the AnomalyCLIP checkpoint 9_12_4_multiscale/epoch_15.pth and CLIP ViT-L/14@336px. Images are re-sized and center-cropped to 518×518 with OpenAI CLIP normalization. Cached features are extracted from layers $\{6, 12, 18, 24\}$, while controlled WPTA ablations aggregate feature map layers $\{1, 2, 3\}$. Layer fusion uses summation, Gaussian smoothing uses $\sigma = 5$, and AUPRO uses 200 thresholds.

Prototype adaptation uses temperature 0.07, semantic sharpening $\gamma = 1.0$, wavelet evidence exponent $\eta = 1.0$, and evidence top- k ratio 0.2. Boundary-aware wavelet reliability uses boundary-aware mode with mix weight 0.05. Except for the fixed-baseline row, prototype/fusion variants in controlled ablations share the same multi-crop and pixel-to-image settings.

4.2 Results

Industrial Results Table 1 follows the compact multi-method layout of the referenced LNCS paper. It reports representative CLIP-based ZSAD methods together with our current fixed AnomalyCLIP baseline and WPTA results. External rows are used as protocol-reference positioning; the same-protocol claim remains the improvement over the current fixed AnomalyCLIP baseline and the controlled ablations below.

Table 1. Zero-shot anomaly detection results on five industrial benchmarks. Each cell reports paired metrics. Best and second-best values among collected numbers are marked in **bold** and underline. [†] denotes reproduced results in the referenced table, and [‡] denotes our current fixed-baseline evaluation.

Image-level results (Image-AUROC, Image-AP)									
Dataset	CLIP	WinCLIP	VAND	CoOp	AdaCLIP	AnomalyCLIP [†]	AA-CLIP [†]	TAAP/INPC	WPTA
MVTec AD	(74.1, 87.6)	(91.8, 96.5)	(86.1, 93.5)	(88.8, 94.8)	(89.2, 96.4)	(91.6, 96.4)	(90.5, 94.9)	(<u>92.8</u> , <u>96.7</u>)	(94.5 , 97.6)
VisA	(66.4, 71.5)	(78.1, 81.2)	(78.0, 81.4)	(62.8, 68.1)	(85.8 , 84.9)	(82.0, 85.3)	(<u>84.6</u> , 82.2)	(83.3, <u>85.9</u>)	(<u>84.6</u> , 87.4)
MPDD	(54.3, 65.4)	(63.6, 69.9)	(73.0, 80.2)	(55.1, 64.2)	(76.0, 80.4)	(73.7, 76.5)	(75.1, 80.1)	(81.8 , 84.8)	(<u>77.8</u> , <u>82.3</u>)
BTAD	(34.5, 52.5)	(68.2, 70.9)	(73.6, 68.6)	(66.8, 77.4)	(88.6, 92.4)	(89.1, 91.0)	(94.8 , 97.9)	(93.3, 93.1)	(<u>93.9</u> , <u>94.9</u>)
DTD-Synthetic	(71.6, 85.7)	(93.2, 92.6)	(86.4, 95.0)	(-, -)	(95.5, 97.0)	(94.5, 97.7)	(93.3, 97.8)	(97.5 , 99.0)	(<u>96.9</u> , <u>98.7</u>)
Average	(60.2, 72.5)	(79.0, 82.2)	(79.4, 83.7)	(68.4, 76.1)	(87.0, 90.2)	(86.2, 89.4)	(87.7, 90.6)	(89.7 , <u>91.9</u>)	(<u>89.5</u> , 92.2)
Pixel-level results (Pixel-AUROC, Pixel-AUPRO)									
MVTec AD	(38.4, 11.3)	(85.1, 64.6)	(87.6, 44.0)	(33.3, 6.7)	(88.7, 37.8)	(91.2, 83.2)	(91.9 , 84.6)	(91.5, <u>85.5</u>)	(<u>91.8</u> , 85.6)
VisA	(46.6, 14.8)	(79.6, 56.8)	(94.2, 86.8)	(24.2, 3.8)	(95.5, 56.8)	(95.5, 86.7)	(95.5, 83.0)	(<u>95.6</u> , <u>88.3</u>)	(96.2 , 91.3)
MPDD	(62.1, 33.0)	(76.4, 48.9)	(94.1, 83.2)	(15.4, 2.3)	(96.1, 60.3)	(<u>96.9</u> , 84.6)	(96.7, 76.5)	(96.7, 89.9)	(97.3 , 89.9)
BTAD	(30.6, 4.4)	(72.7, 27.3)	(60.8, 25.0)	(28.6, 3.8)	(92.1, 32.5)	(93.5, 70.5)	(97.0 , 69.0)	(94.1, <u>77.1</u>)	(<u>96.3</u> , 78.2)
DTD-Synthetic	(33.9, 12.5)	(83.9, 57.8)	(95.3, 86.9)	(-, -)	(97.7, 75.0)	(97.4, 89.1)	(96.4, 85.9)	(97.4, <u>91.5</u>)	(97.9 , 91.8)
Average	(42.3, 15.2)	(79.5, 51.1)	(86.4, 65.2)	(25.4, 4.2)	(94.0, 52.5)	(94.9, 82.8)	(<u>95.5</u> , 79.8)	(95.1, <u>86.5</u>)	(95.9 , 87.4)

Core Component Ablation Table 2 validates the WPTA mechanism on MVTEC and VisA. Direct wavelet fusion is worse than the fixed baseline, showing that wavelet responses should not be used as final anomaly scores. Semantic prototype adaptation improves over fixed prototypes, while wavelet-guided adaptation and conservative update further improve Pixel-AUPRO.

Table 2. Core component ablation on MVTec and VisA. Each cell reports (P-AUROC, P-AUPRO) / (I-AUROC, I-AP).

Method	MVTec AD	VisA
Baseline	(91.2, 83.2) / (91.6, 96.4)	(95.5, 86.7) / (82.0, 85.3)
Direct wavelet fusion / no adaptation	(88.7, 80.4) / (92.9, 96.9)	(94.6, 85.1) / (81.6, 84.8)
Semantic prototype adaptation	(91.6, 85.2) / (93.7, 97.1)	(96.0, 90.4) / (83.7, 86.9)
Wavelet prototype adaptation w/o conservative	(91.7, 85.8) / (93.9, 97.2)	(96.1, 91.3) / (84.1, 87.0)
Full WPTA controlled setting	(91.8, 86.2) / (94.1, 97.4)	(96.2, 91.7) / (84.3, 87.3)

Wavelet Design Ablation Table 3 studies the role of wavelet reliability. HF-only reliability gives limited gains, while boundary-aware reliability improves Pixel-AUPRO by suppressing structure-boundary pseudo evidence.

Table 3. Wavelet reliability design ablation. Each cell reports (P-AUROC, P-AUPRO) / (I-AUROC, I-AP).

Wavelet setting	MVTec AD	VisA
Semantic-only prototype adaptation	(91.6, 85.2) / (93.7, 97.1)	(96.0, 90.4) / (83.7, 86.9)
Direct wavelet fusion	(88.7, 80.4) / (92.9, 96.9)	(94.6, 85.1) / (81.6, 84.8)
HF-only W + prototype adaptation	(91.6, 85.3) / (94.0, 97.2)	(96.0, 90.8) / (84.0, 86.9)
Boundary-aware W + prototype adaptation	(91.7, 85.7) / (93.8, 97.3)	(96.1, 91.2) / (83.9, 87.1)
Full boundary-aware W + conservative	(91.8, 86.2) / (94.1, 97.4)	(96.2, 91.7) / (84.3, 87.3)

Configuration Audit

Table 4 records which modules are active in each dataset-specific final system. It prevents over-attributing all five-dataset gains to the WPTA mechanism.

Table 4. Configuration audit for the five-dataset calibrated inference system.

Dataset	Result	Wavelet reliability	Prototype rectification	Multi-crop	Pixel-to-image
MVTec AD	91.8 / 85.6 / 94.5 / 97.6	yes	yes	yes	yes
VisA	96.2 / 91.3 / 84.6 / 87.4	yes	yes	yes	yes
MPDD	97.3 / 89.9 / 77.8 / 82.3	no	no	yes	yes
BTAD	96.3 / 78.2 / 93.9 / 94.9	no	no	yes	yes
DTD-Synthetic	97.9 / 91.8 / 96.9 / 98.7	yes	no	yes	yes

5 Discussion

Why wavelet reliability, not wavelet scoring.

Wavelets provide local frequency structure rather than semantic anomaly decisions. The direct-fusion negative control shows that adding wavelet response to

the anomaly map hurts localization. WPTA instead uses wavelets only as evidence reliability and lets calibrated CLIP prototypes produce the final anomaly map.

Why conservative calibration.

Unlabeled test-time calibration can suffer from prototype drift. WPTA uses confidence-gated conservative updates so that visual anchors affect prototypes only when the evidence is reliable. In the default setting, the abnormal prototype update strength is zero, and abnormal evidence mainly controls the update gate. This makes the method conservative rather than a large bidirectional prototype shift.

Scope and limitations.

The five-dataset final systems use dataset-specific configurations and therefore support system-level gains over the fixed AnomalyCLIP baseline, not a causal claim that WPTA is isolated on all five datasets. The WPTA mechanism is validated by controlled MVTec/VisA ablations. External method comparisons require careful protocol alignment before being used for ranking claims.

6 Conclusion

We proposed WPTA for CLIP-based zero-shot anomaly detection. The central idea is to use Haar wavelet cues as patch evidence reliability for test-time prototype adaptation instead of direct final-map fusion. Controlled MVTec/VisA ablations show that direct wavelet fusion hurts localization, while semantic-spectral evidence selection and conservative prototype calibration improve test-time adaptation. Across five industrial benchmarks, the calibrated inference system improves the fixed AnomalyCLIP baseline, especially on Pixel-AUPRO. These results support a conservative conclusion: wavelet cues are more useful as evidence reliability than as anomaly scores, and test-time prototype calibration can reduce the mismatch between fixed CLIP prototypes and instance-specific defect appearances.

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